sketch_jul16d | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

sketch_jul16d.ino

```
int redled=12;
int time=1000;

void setup() {
  // put your setup code here, to run once:
pinMode(redled,OUTPUT);
pinMode(LED_BUILTIN,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(redled,HIGH);
digitalWrite(LED_BUILTIN,LOW);
delay(1000);

digitalWrite(redled,LOW);
digitalWrite(LED_BUILTIN,HIGH);
delay(1000);
}
```

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```
int redled=12;
int time=1000;

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}

void loop() {
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digitalWrite(redled,HIGH);
digitalWrite(LED_BUILTIN,LOW);
delay(time);

digitalWrite(redled,LOW);
digitalWrite(LED_BUILTIN,HIGH);
delay(time);
}

```
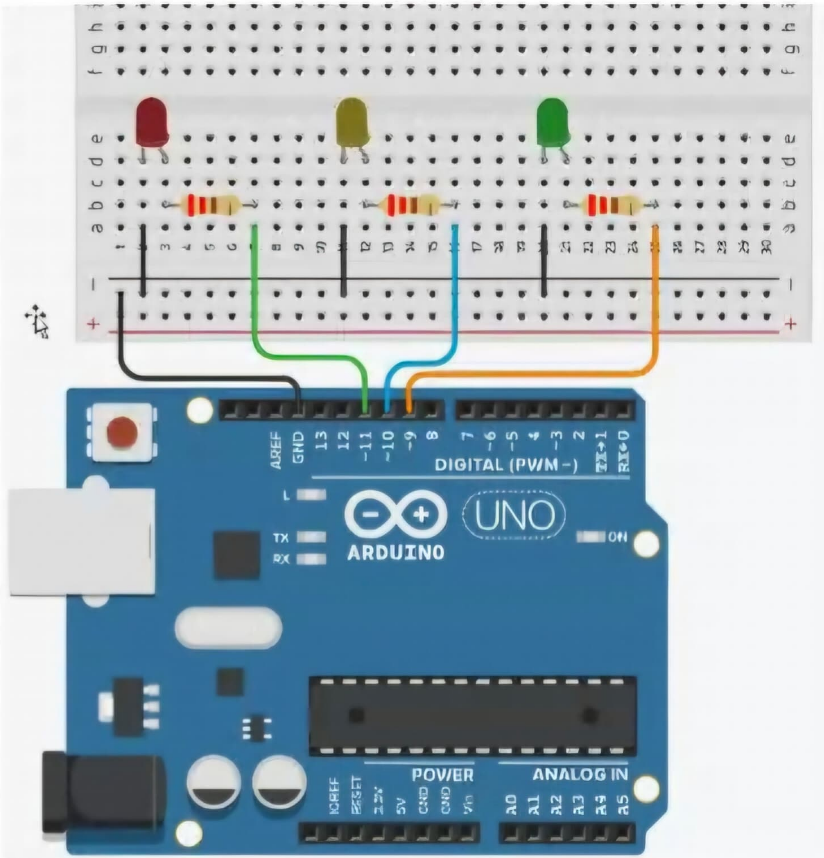
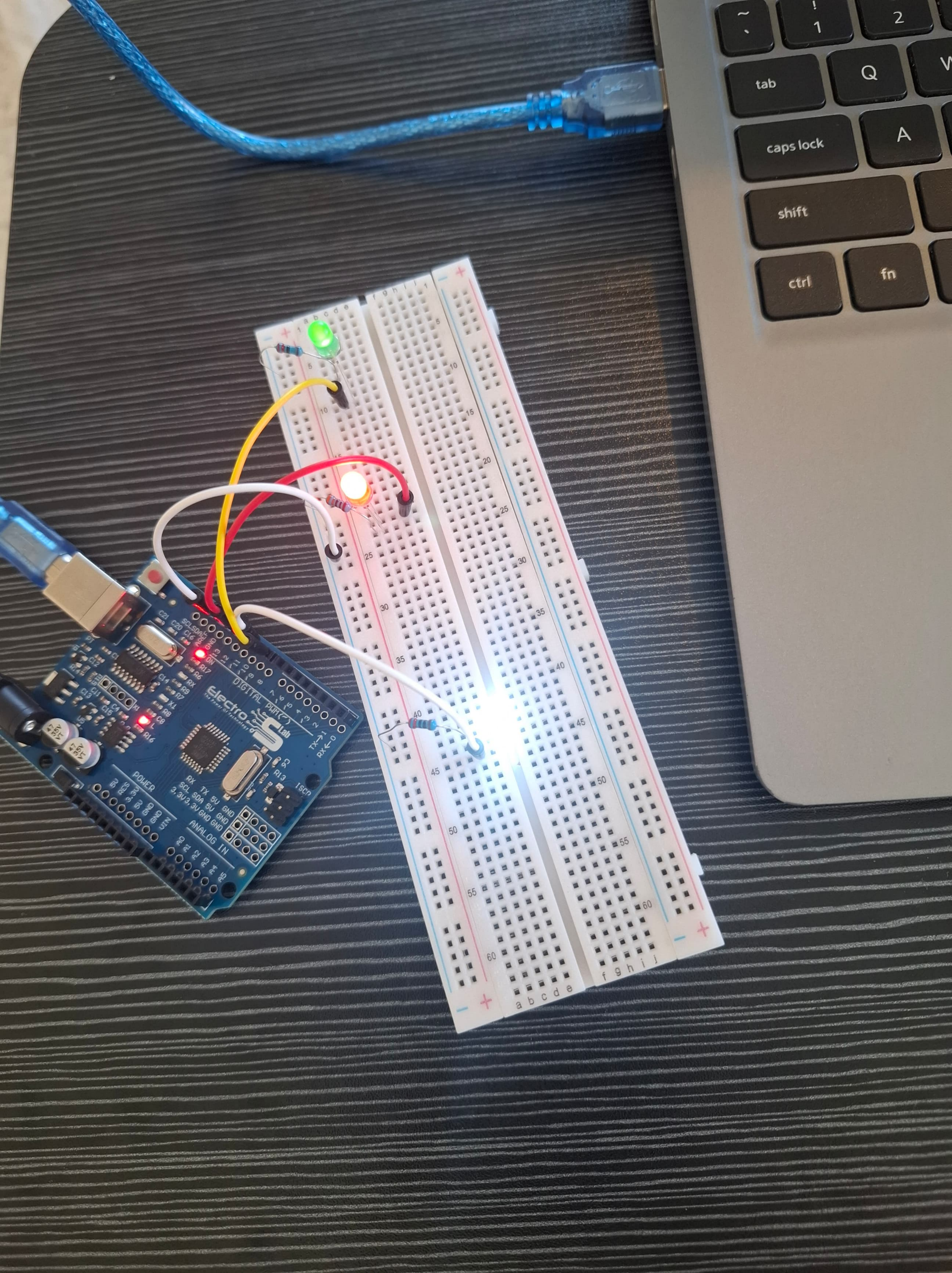

Board connection



RGW_LED.ino

```
int time=1000;
int redled=12;
int whiteled=9;
int greenled=8;

void setup() {
  // put your setup code here, to run once:
pinMode(redled,OUTPUT);
pinMode(whiteled,OUTPUT);
pinMode(greenled,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(redled,HIGH);
digitalWrite(whiteled,HIGH);
digitalWrite(greenled,HIGH);
delay(time);

digitalWrite(redled,LOW);
digitalWrite(whiteled,LOW);
digitalWrite(greenled,LOW);
delay(time);
}
```


tab
Q
caps lock
A
shift
ctrl
fn
1
2
Electro
DIGITAL PWM(~)
POWER
ANALOG IN

RGB LED

The RGB Light is a compound LED that has the main three lights: red, green, and blue.

The RGB consists of three lights, each one has its own pin. We can turn ON the needed color by sending electricity at the right pin:

1- We can have one of the main colors

2- We can have a white color if we turn ON the three pins

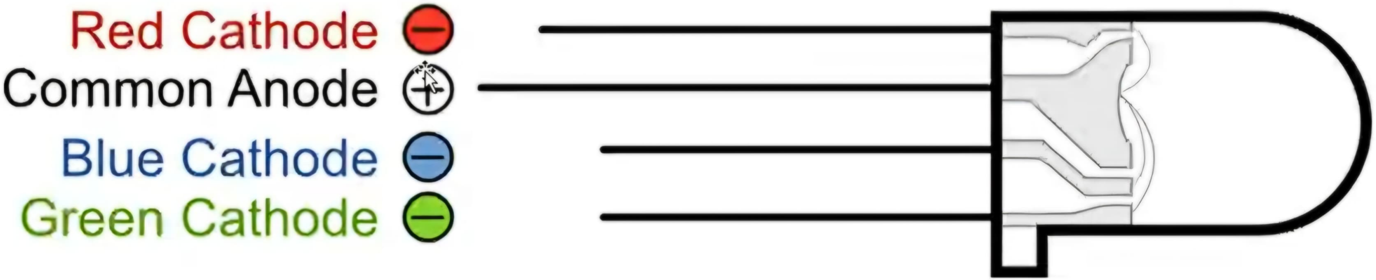
3- We can have a different color if we combine different colors together

Pin Types for RGB LED

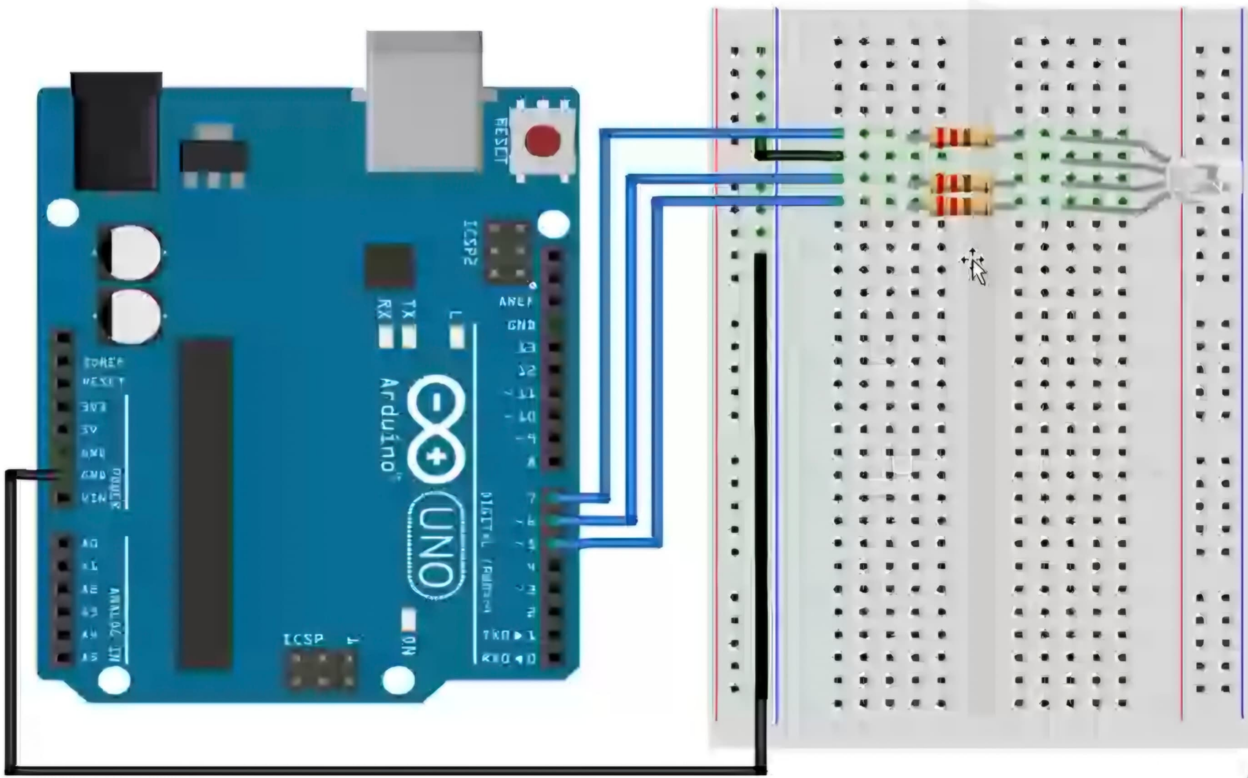
The RGB LED rope has 4 pins:

1. One is for VCC,
2. 3 is for the GND red, blue, green

Depending on the signal frequency, the LEDs will glow and change color.



Board connection



dot

Arduino UNO

sketch_jul16h.ino

```
digitalWrite(red,HIGH);//RED+GREEN=YELLOW
digitalWrite(green,HIGH);
delay(time);
digitalWrite(red,LOW);
digitalWrite(green,LOW);

digitalWrite(green,HIGH);//GREEN+BLUE=LIGHT BLUE
digitalWrite(blue,HIGH);
delay(time);
digitalWrite(green,LOW);
digitalWrite(blue,LOW);

digitalWrite(red,HIGH); //RED+GREEN+BLUE=WHITE
digitalWrite(blue,HIGH);
digitalWrite(green,HIGH);
delay(time);
digitalWrite(red,LOW);
digitalWrite(blue,LOW);
digitalWrite(green,LOW);
}
```

RGB.ino

```
int red=7;
int green=6;
int blue=5;
int time=1000;

void setup() {
  // put your setup code here, to run once:
pinMode(red,OUTPUT);
pinMode(green,OUTPUT);
pinMode(blue,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(red,HIGH);
delay(time);
digitalWrite(red,LOW);

digitalWrite(green,HIGH);
delay(time);
digitalWrite(green,LOW);

digitalWrite(blue,HIGH);
delay(time);
digitalWrite(blue,LOW);

digitalWrite(red,HIGH);
digitalWrite(green,HIGH);
delay(time);
digitalWrite(red,LOW);
digitalWrite(green,LOW);
delay(time);
```